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Question Paper Code: E3115

B.E./B.Tech. DEGREE EXAMINATION, APRIL/MAY 2010

Third Semester

Information Technology

IT2201 — DATA STRUCTURES AND ALGORITHMS

(Regulation 2008)

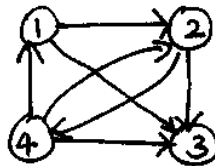
Time: Three hours

Maximum: 100 Marks

Answer ALL Questions

PART A — (10 × 2 = 20 Marks)

1. Why do we need ADT for implementing Data Structures?
2. Write down a function to insert an element into a queue, in which the queue is implemented as an array.
3. Represent the following in a AVL tree
k, t, e, v, p, a, m, u, h.
4. Write down pseudocode to perform postorder traversal of a binary tree.
5. Why do we need a Hash table as a data structure as compared to any other data structure?
6. What are the methods for avoiding collision?
7. What is topological sorting?
8. Represent the following graph as an adjacency matrix



9. Prove that $(n + n \lg n) = O(n \log n)$ is true or false.
10. Solve the recurrence equation $T(n) = 2T(n-1) + 1, T(1) = 1$.

PART B — (5 × 16 = 80 Marks)

11. (a) Write a C program to perform the following stack operations
- Declare a structure with one integer data type. Add necessary variables to ensure that you can create a linked representation.
- (i) Write a function called 'push' that takes two parameters: An integer variable and a stack into which it would push this element and returns a 1 or a 0 to show success of addition or failure. (6)
 - (ii) Write a function called 'pop' that takes two parameters: An integer pointer and a stack from which we need to pop this element. (6)
 - (iii) A display function that would print the contents of stack. Add necessary functions to check for emptiness or fullness of the stack. (4)

Or

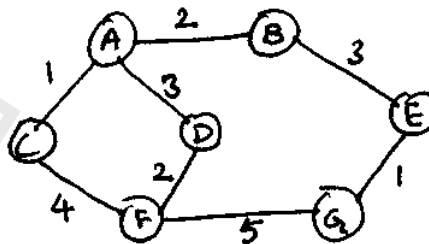
- (b) Write algorithms to perform the following in a doubly linked list:
- (i) To insert an element in the beginning, middle, end of the list (8)
 - (ii) To delete an element from anywhere in the list (8)

An element is a structure variable that contains an integer data field and a string data field.

12. (a) (i) Show that the maximum number of nodes in a binary tree of height H is $2^{H+1} - 1$. (4)
- (ii) Construct an expression tree for the expression $(a + b * c) + (d * e / f) - g$. (4)
- (iii) Write pseudocode to insert and delete from a binary search tree. (8)

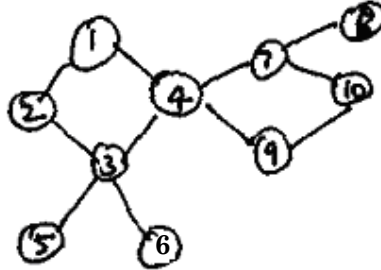
Or

- (b) (i) Show the result of inserting the nodes 2, 1, 4, 5, 9, 3, 6, 7 into an initially empty Binary Search tree. (6)
- (ii) Write down the functions to insert into an AVL tree by calling appropriate functions to perform double and single rotations also. (10)
13. (a) (i) Given the input {4371, 1323, 6173, 4199, 4344, 9679, 1989} and a hash function of $h(X) = X \pmod{10}$ show the resulting
- (1) Separate Chaining hash table (5)
- (2) Open Addressing hash table using linear probing (5)
- (ii) What are the advantages and disadvantages of various collision resolution strategies? (6)
- Or
- (b) (i) State and explain the dynamic equivalence problem. (6)
- (ii) Explain the path compression algorithm and analyse the Union/Find algorithm used. (10)
14. (a) (i) Give the Prim's algorithm for determining minimum spanning tree and analyse this algorithm for its worst case time complexity. (10)
- (ii) Determine the minimum spanning tree for the following graph using Prim's algorithm. (6)



Or

- (b) (i) Write an algorithm to determine the biconnected components in the given graph. (10)
- (ii) Determine the biconnected components in the following graph. (6)



15. (a) What is a greedy algorithm? Give two examples that can be solved using greedy algorithm and explain them with example. (16)

Or

- (b) What is NP completeness? Explain an NP complete problem with example. (16)